# Group 6 Study Notes and Reasoning

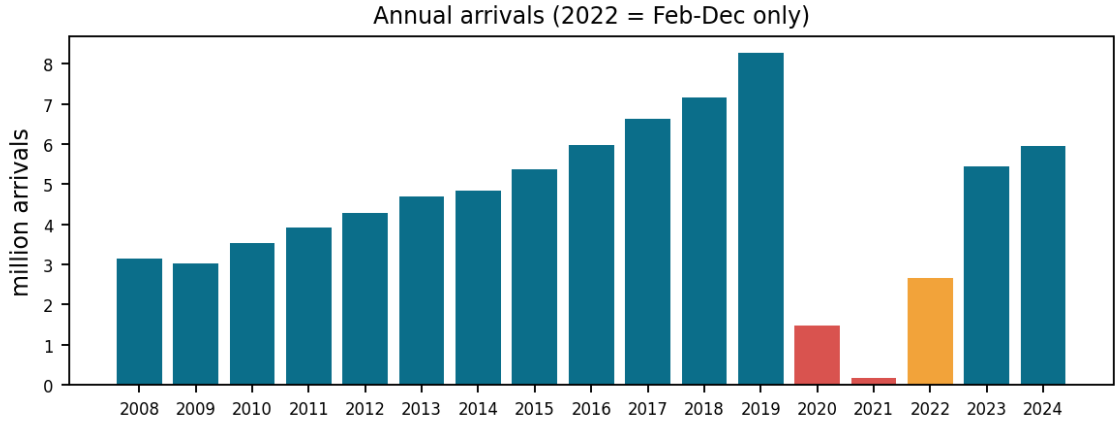
Philippine tourism: visitor arrivals by market, 2008-2024 | DS311 Exploratory Data Analysis, Midterm Lab

## 1. The story in one page

**Question.** How fast has international tourism to the Philippines grown since 2008, which source markets drove it, and how did COVID-19 change how markets move relative to each other?

**Answer.** Arrivals grew about 9.2% a year from 3.14 million (2008) to 8.26 million (2019). COVID-19 was a structural break (monthly arrivals fell 99.9% from January to April 2020). By 2024 arrivals were 5.95 million, 72% of 2019. China explains about half of the remaining gap; Korea and the USA now hold a bigger share than before. Markets look strongly correlated in levels, but most of that is a shared upward trend, not month-to-month co-movement. The year-end/winter peak (January and December) and the September-October low are the seasonal pattern.

| Headline number | Value | Where it comes from |
|---|---|---|
| Arrivals 2008 / 2019 | 3,139,422 / 8,260,913 | Sum of the 12 monthly grand totals in the 2008 and 2019 PDFs |
| CAGR 2008-2019 | 9.2% | $(8{,}260{,}913 / 3{,}139{,}422)^{1/11} - 1$ |
| 2020 / 2021 change | -82.1% / -88.9% | 1,482,535 and 163,879 arrivals |
| Lowest month | 948 (Apr 2020) | -99.9% vs Jan 2020 (796,164) |
| Full-year 2023 / 2024 | 5,450,557 / 5,949,350 | Share of 2019: 66% and 72% |
| China share, Jan-Oct | 22.1% (2019) to 5.0% (2023) to 5.8% (2024) | Share of all arrivals |
| Gap to 2019 (Jan-Oct 2023, 50 markets) | 2,655,646 | China 48.1%, Korea 16.5%, Japan 12.1%, USA 4.9%, Taiwan 4.4% |
| Seasonal peak / trough | Jan 115.8, Dec 115.5 / Sep 82.5 | Trend-adjusted index, 100 = average month |




## 2. Data: what we have and why it needed care

The repository holds the **17 original DOT yearly PDF reports** (2008-2024), not a spreadsheet. A PDF is a picture of a table, so extraction is itself a data-quality problem. The reasoning below is what a reviewer will ask about.

### 2.1 Four layouts, so four readers

| Years | Table | Months | Basis |
|---|---|---|---|
| 2008-2021 | Visitor arrivals by country of residence | Jan-Dec | Foreign visitors by residence; "Overseas Filipinos" on a separate row |
| 2022 | Air visitor arrivals by nationality | Feb-Dec | Nationality, air only |
| 2023 | Ranking by country of residence (also a nationality table) | Jan-Oct | Residence; overseas Filipinos separate |
| 2024 | Top-60 markets (a citizenship table and a residence table) | Jan-Dec | Residence; no overseas-Filipino row |

### 2.2 Extraction and proof that it is right

- **Split-number bug.** The default PDF reader returns `179,900` as `1` and `79,900`. This silently corrupts sums. Fix: rebuild each line from individual characters and start a new value only after a visible gap.
- **Reconciliation.** Every PDF prints subtotals and a grand total. If the extracted country rows add up to them, the extraction is correct. 2009-2021 match exactly (gap 0); 2008 differs by at most 28 in a month, 2022 by 11, 2023 by 105; 2024 adds up exactly. The 2019 total equals the official 8,260,913.
- **Independent check.** Pre-COVID statistics for Korea, USA, China and Japan are identical to those from an earlier compiled CSV of the same DOT data.

### 2.3 Design decisions and their reasons

| Decision | Reason |
|---|---|
| Core panel of 50 markets | A market can only be followed 2008-2024 if it is listed in every year 2008-2021 *and* in the 2024 top-60. The remainder is kept as other_untracked. |
| Jan 2022 and Nov-Dec 2023 left as NaN | They are not reported, not zero. Zero would create fake collapses. No interpolation; rolling windows touching a gap stay blank. |
| Full-year 2023 = 5,450,557 | Printed in the 2024 PDF. Implies 1,005,277 for Nov+Dec 2023 combined, which cannot be split by month. |
| Dash "-" set to 0 and flagged (was_dash) | DOT uses a dash for no recorded arrivals. 47 cells in 27 markets (28 in 2020, 3 in 2021, 16 in 2022). Flag keeps the assumption reversible. 2022 dashes in larger markets (China, Taiwan, India...) may be suppressed values, so treat with caution. |
| 2022 kept but flagged (basis) | Nationality/air basis. For Jan-Oct 2023 the nationality and residence tables are close (Korea 1,178,292 vs 1,170,498). |
| Market comparisons with 2019 use 2023 as main benchmark | The 2024 table has no overseas-Filipino row and its rows sum exactly to the grand total. Check: the Nov-Dec 2023 figure implied by the two PDFs is 0.94 of Nov-Dec 2024 for the grand total (sensible) but 1.13 to 1.66 for every individual market (not plausible growth). Likely a definition break, so 2024 market levels are indicative. |
| Common window Jan-Oct for market recovery | The 2023 report stops in October. Comparing different month sets would mix seasonality into the recovery figure. |

## 3. Method notes: what we did and why

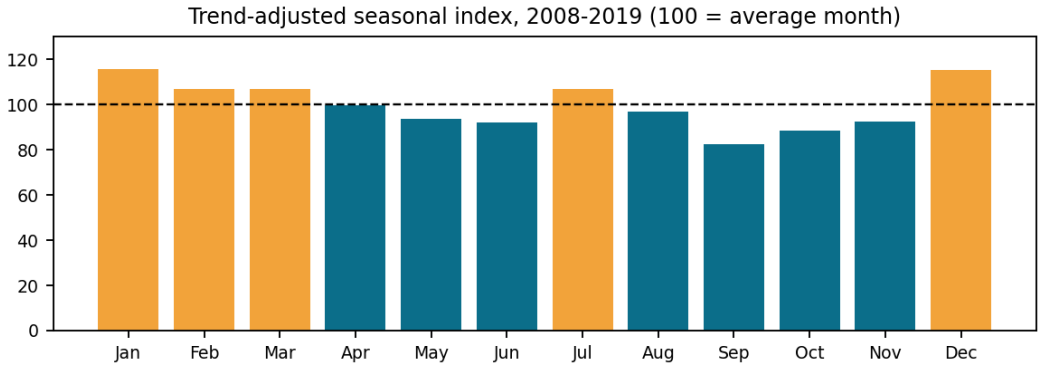
### 3.1 Univariate: distributions (Section 4)

- **Pre-COVID split (2008-2019, n = 144).** COVID is a structural break; one shock would dominate the variance. Coefficient of variation rises from 33% to 49% when all 201 months are included.
- **Skewness and excess kurtosis.** Total: skew 0.60, kurtosis -0.50 (right-skewed, flat). Reason: growth pushes later months into the upper tail. China is the most volatile (CV 86%, skew 1.32) because it grew from a small base.

- **Shapiro-Wilk.** Rejects normality for all five variables ($p < 0.05$). Expected: trending, seasonal series are not bell-shaped.
- **IQR rule.** Flags 7 months (USA Jan 2018, Jan 2019, Dec 2019; China Feb, Jul, Aug 2019; Japan Aug 2019). All are genuine 2018-2019 peaks, so none is removed.
- **Log transform.** Reduces |skew| for four of five variables, not Korea (about 0.35 to -0.51). Only USA passes Shapiro after the log ($p = 0.14$). The log is used in Section 6 to straighten and stabilise, not to "make data normal".

### 3.2 Growth, composition and seasonality (Section 5)

- **CAGR** = $(\text{end/start})^{1/\text{years}} - 1$. Used for 2008-2019 only; the COVID years would make a single rate meaningless. China 24.0% a year, Korea 11.3%, Japan 6.0%, USA 5.7%.
- **Recovery** = later period divided by the same months of 2019. Jan-Oct 2023: USA 85.1%, Korea 72.7%, Japan 43.5%, China 14.9% of 2019.
- **Composition.** East Asia 47.6% (2008) to 60.1% (2019); Americas 23.7% to 16.4%. The 2021 bars are an artefact of a tiny total (163,879), so do not read them as a real shift.
- **Seasonality: naive vs trend-adjusted.** Arrivals trend up, so late-year months look high from growth alone. Dividing each month by a centred 2x12 moving average removes the trend. Naive: December highest (118.4). Adjusted: January 115.8, December 115.5 almost tied. Weak Sep-Nov coincides with the rainy and typhoon season, but the data cannot prove the cause.




### 3.3 Bivariate: two numerical variables (Section 6)

| Tool | What it does | Result and reading |
|---|---|---|
| Pearson vs Spearman | Pearson measures linear association; Spearman is rank-based and resists extreme values | Level correlations high (median 0.75). Spearman agrees but does **not** protect against a shared trend. |
| Year-over-year log growth | Removes trend and most seasonality | Median correlation falls to 0.10: markets grew in the same decade, not month to month. |
| Fisher z interval | Confidence interval for r | Korea-China pre-COVID r = 0.829 (95% CI 0.77 to 0.87). |
| Linear regression | China on Korea | China = -37,609 + 0.872 x Korea, $R^2 = 0.69$. Predicts negative arrivals below about 43,000 Korean arrivals, so it is not valid there. |
| Log-log regression | Elasticity: % change in China per 1% in Korea | 1.60, $R^2 = 0.73$. Consistent with China growing faster (24.0% vs 11.3%). |
| Durbin-Watson and Newey-West | Detect autocorrelated residuals; robust standard errors | DW = 0.62 (far below 2), so the plain CI (1.44 to 1.76) is too narrow. Robust CI is 1.23 to 1.97. |

| Tool | What it does | Result and reading |
|---|---|---|
| Cook's distance | Finds influential points (threshold 4/n = 0.028) | Oct 2009 (0.30) and Sep 2009 (0.24). Dropping 5 months moves elasticity from 1.60 to 1.79: report "about 1.2 to 2". |
| Detrending | Removes each series linear time trend | Correlation drops from 0.85 to 0.14: most of the link is a shared trend. |
| Correlation by period | Same pairs in three eras | Pre-COVID 0.61-0.95; COVID near 1 (two clusters, one shock); recovery (n = 33) 0.66-0.95, driven by the common reopening trend. |
| Rolling 12-month correlation | Shows stability over time | Pre-COVID range -0.36 to 0.81, median 0.40, never reaches 0.83. Recovery windows fall from 0.95 to 0.61 during 2023; the 2024 window is 0.49. |




## 4. Likely questions and short answers

**Q. Why use PDFs and not a CSV?**

The repository stores the official DOT reports. They also add 2022, late 2023 and 2024 and the official totals. The cost is four layouts, handled in Section 3.

**Q. How do you know the PDF numbers are right?**

Rows reconcile to the printed totals (gap 0 for 2009-2021), the 2019 total equals the official 8,260,913, and the pre-COVID market statistics match an earlier independent CSV.

**Q. Why not treat the missing months as zero?**

They are unreported. Zero would invent a second collapse. They stay NaN and nothing is interpolated.

**Q. Why compare Jan-Oct and not full years?**

The 2023 report ends in October. The same months in both years keep seasonality out of the comparison. Full-year comparisons use the DOT totals, available for 2019, 2023 and 2024.

**Q. Why is the 2024 market data "indicative"?**

The 2024 table has no overseas-Filipino row and its market rows look broader than the 2023 rows (market ratios 1.13-1.66 against 0.94 for the total). The total is consistent; market levels may be somewhat high, especially USA, Canada, Japan, Australia.

**Q. Is Korea-China r = 0.83 a real relationship?**

Only partly. After removing each series linear trend it is 0.14, and rolling correlations inside a year are much lower. Most of it is shared growth.

**Q. Why are the COVID and recovery correlations high?**

COVID: two clusters (before and after the collapse) make r near 1. Recovery: every market climbed out of the border closures together. In 2024, once that trend fades, the Korea-China rolling r is 0.49.

**Q. Why log transform if the data are still not normal?**

To straighten the relationship and stabilise variance for regression, not to achieve normality.

**Q. What would you do with more data?**

Spending, length of stay, purpose of visit, and a consistent residence-based series for 2022 and 2024 so the basis breaks disappear.

## 5. Limitations to state openly

- Four PDF layouts; 2022 is air arrivals by nationality for February-December; January 2022 and November-December 2023 are absent; 2024 lists only the top 60 markets.
- The 50 core markets cover about 97% of the grand total in 2019; the 2024 market rows appear broader than earlier years.
- Overseas Filipinos are separate, and their share changes (0.9% in 2019, 23.7% in 2022, 8.3% in Jan-Oct 2023), so the grand total mixes groups in changing proportions.
- Dashes were treated as zero (47 cells, 16 in 2022 including some larger markets).
- Monthly series are trending and autocorrelated, so p-values and intervals are descriptive; the robust interval and the detrended correlation are the honest summaries.
- Arrivals only: no spending, length of stay or purpose of visit.

Source: Philippine Department of Tourism visitor arrival statistics (remote.tourism.gov.ph/tourism/demand-statistics); PDFs in github.com/Ajooooooo/EDA-Group-6, data/raw. Figures in these notes come from Group6_Midterm_Lab_Notebook.ipynb.